Data dictionary

CSV files are UTF-8 with headers. Identifiers and MCC codes are strings. Timestamps use `YYYY-MM-DD HH:MM:SS` on a common fictional clock; no timezone conversion is needed. Cutoff: 2026-01-01 00:00:00, exclusive. Empty fields mean missing, not zero.

transactions.csv - one completed positive purchase

Each row represents one completed purchase with a positive transaction value.

Column	Type	Meaning
transaction_id	String	Unique purchase identifier.
user_id	String	Purchaser; joins to users.csv.
merchant_id	String	Merchant receiving payment; joins to merchants.csv.
payment_source_id	String	Card used; joins to cards.csv.
transaction_timestamp	Timestamp	Purchase completion time.
volume_nok	Decimal	Purchase value expressed in NOK.
revenue_nok	Decimal	NOK amount the fictional payment service retains: a merchant-specific percentage of volume_nok, rounded to two decimals. Before costs; not profit.
currency	String	NOK, regardless of user or merchant country.
user_country	String	Norway or Denmark, repeated from users.csv for convenience. Not physical location at payment time.
payment_channel	String	`online` or `in_person`, a broad channel rather than device or product feature.

Fee percentages are synthetic, fixed per merchant during the year, and do not represent actual Vipps fees. NOK values are synthetically rescaled, not converted at a historical exchange rate. International purchases are already in NOK. Failed payments, refunds and transfers are excluded.

users.csv - one user

Column	Type	Meaning
user_id	String	Unique user identifier.
observation_start_date	Timestamp	Start of complete purchase coverage, not signup or first use.
vip_enrolled	Boolean	True if enrolled in VIP by the dataset cutoff; false otherwise. Not membership status at every historical purchase.
vip_enrolled_at	Timestamp, nullable	VIP signup time; null for non-members at cutoff. No membership cancellations are modelled.
user_country	String	Synthetic residence: Norway or Denmark. Authoritative country attribute.
age_bucket	String	Synthetic age as of 1 January 2025: four-year buckets `18-21`, `22-25`, …, `78-81`.

All users share the cutoff. Behaviour before observation_start_date is unknown. The selected population is not representative of actual customers. Country and age are generated for the exercise.

merchants.csv - one merchant

Column	Type	Meaning
merchant_id	String	Unique merchant identifier.
merchant_name	String	Fictional grocery, specialist-food or online retailer.
merchant_country	String	Norway, Denmark, Sweden, Germany or Netherlands.
merchant_mcc_code	String, nullable	MCC category; joins to mcc.csv when known. Missing means unclassified, not “other”.

Users can purchase from international merchants. Grocers support online and in-person payments; non-food retailers accept online payments. Merchant country is not the transaction currency. Resemblance of fictional names to actual businesses is incidental.

mcc.csv - one category

Column	Type	Meaning
merchant_mcc_code	String	Unique Merchant Category Code (MCC).
category_name	String	Category description.

This is MCC, not NACE. Several merchants share a category. Join transactions → merchants → mcc; preserve transaction grain in both joins.

cards.csv - one card/payment source (optional)

Column	Type	Meaning
payment_source_id	String	Unique card identifier used in transactions.
user_id	String	Card owner. A user can have several cards.
card_type	String	Synthetic `debit` or `credit`.
card_status	String	`Active` or `Inactive` as of 1 January 2026.
card_scheme	String	`visa` or `mastercard`, the network/scheme. Issuing bank is not supplied.

Use this optional table only if it supports your chosen question. You do not need to use every supplied table.

No card numbers, expiry dates or security codes are included. Cutoff status does not describe status at the historical purchase time: an inactive card can have valid earlier purchases. Status history is unavailable. Check purchase-card ownership using payment_source_id and user_id; joining transactions to all of a user's cards would multiply purchases.

experiment_exposure.csv - one allocation per user

Column	Type	Meaning
user_id	String	Eligible non-member allocated to the invitation experiment.
experiment_id	String	`vip_invitation_v1`, the only experiment.
variant	String	`control` (no invitation) or `invitation` (assigned a VIP invitation). Not actual membership.
exposure_timestamp	Timestamp	Randomised assignment/start time for either group; not observed interaction or VIP signup.

The brief describes eligibility, randomisation and waves. Allocation applies across all cards and merchants. Control users can join organically; invited users can decline. Use vip_enrolled_at when assessing behaviour before/after membership. Users outside the trial can also be VIP members. Purchases or signups after cutoff are unobserved. No invitation interaction events or block identifiers are supplied.